Sampling in Astrophotography

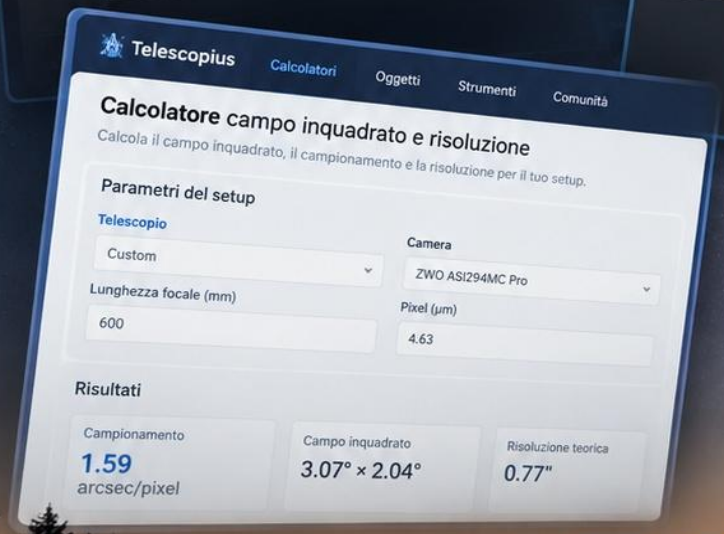
# CAMPIONAMENTO IN ASTROFOTOGRAFIA

Seeing · FWHM · Nyquist · Pixel · BIN












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Deep Sky Astrophotography from Italy

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The document follows a progressive path: from seeing and FWHM to sampling, BINNING, and practical application to the telescope + camera system.

Objective: understand the concept starting from the actual sky at the observing location and arrive, without logical jumps, at practical use with the ASI294MC Pro, GSO 150/600 and RC8 used as examples.

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# Sampling in Astrophotography

### 1. Where do we start? The sky

When we photograph a star, the star is essentially a point. However, the telescope does not record a perfect point. The light is spread over a small spot because of the atmosphere, the optics, focusing, tracking, vibrations and other factors.

The first thing to distinguish, therefore, is seeing. Seeing is the effect of the atmosphere on resolution: in simplified terms, it indicates how much the atmosphere tends to broaden the image of a point source.

If on a given night we assume a seeing of 2.5”, this does not mean that every star will necessarily have a measured width of exactly 2.5”. It means that the atmosphere represents an important limit to the resolution that can be achieved.

- Seeing belongs to the atmosphere.
- It does not depend on the camera.
- It does not depend on focal length.
- It does not depend on sampling.
- Sampling, instead, determines how finely we record what the system allows us to see.

### 2. The star on the sensor: PSF and FWHM

A star therefore does not appear as a single bright pixel. It appears as a distribution of brightness: the centre is brighter and the brightness decreases away from the centre. This distribution is called the PSF, Point Spread Function.

To describe the width of the star quantitatively, FWHM, Full Width at Half Maximum, is often used; that is, the width of the distribution measured at 50% of the maximum brightness.

Imagine that the peak of the star is 100. We find the two points on either side where the brightness has fallen to 50 and measure the distance between them. That distance is the FWHM.

FWHM is therefore a measure of the apparent width of the star in our image. It can be expressed in pixels or in arcseconds.

### 3. FWHM in pixels and FWHM in arcseconds are not the same thing

This distinction is fundamental. If PixInsight measures FWHM = 3.5 pixels, that value alone does not yet tell us how large the star is on the sky.

To convert pixels into arcseconds, we need to know the sampling.

Example: if the sampling is 1.5”/pixel and a star has an FWHM of 3 pixels, then:

**Angular FWHM = 3 × 1.5 = 4.5”**

Therefore, the FWHM measured in the photograph corresponds to 4.5 arcseconds.

This is the first essential connection:

**FWHM in arcseconds = FWHM in pixels × sampling in arcseconds/pixel**

### 4. What sampling really is

Sampling tells us how many arcseconds of sky are represented by each individual sensor pixel.

It is expressed in arcseconds/pixel (”/pixel).

For example, 1.5”/pixel means that each pixel represents a portion of sky 1.5 arcseconds wide.

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# Sampling in Astrophotography

This is the definition to remember. Sampling does not tell us how good a photograph is, nor does it directly tell us how good the seeing is: it tells us the scale at which the telescope + camera system records the sky.

### 5. What sampling depends on

Dipende principalmente da due elementi: dimensione dei pixel e lunghezza focale.

- Larger pixels → each pixel covers more sky → greater sampling.
- Smaller pixels → each pixel covers less sky → smaller sampling.
- Longer focal length → each pixel covers a smaller portion of sky → smaller sampling.
- Shorter focal length → each pixel covers a larger portion of sky → greater sampling.

The practical formula is:

**Sampling (”/pixel) = 206,265 × pixel size (mm) / focal length (mm)**

If the pixel is expressed in micrometres, the convenient form is:

**Sampling (”/pixel) = 206,265 × pixel (µm) / [focal length (mm) × 1000]**

You do not need to memorise it: it is mainly useful for quantitatively connecting pixel size and focal length to image scale.

### 6. Worked Example

Let’s work through the calculation using an ASI294MC Pro camera, which has physical pixels measuring 4.63 µm, with two telescopes: the GSO 150/600 Newtonian and the Tecnosky GSO RC8 Carbon, a 203 mm Ritchey–Chrétien astrograph with a focal length of 1,624 mm.

| Configuration | Focal length | Sampling |
|---|---|---|
| ASI294 + GSO 150/600 | 600 mm | 1,59”/pixel |
| ASI294 + RC8 | approximately 1624 mm | 0,59”/pixel |

These are two very different setups using the same camera. The 150/600 captures a larger area of sky per pixel, while the RC8 samples a smaller area of sky per pixel.

### 7. Sampling and FWHM: how many pixels describe a star?

Ora possiamo mettere insieme i concetti.

If a star has an angular FWHM of 2.4”:

| Sampling | Pixels per FWHM |
|---|---|
| 2.4”/pixel | 1 pixel |
| 1.2”/pixel | 2 pixel |
| 0.8”/pixel | 3 pixel |
| 0.6”/pixel | 4 pixel |

La relazione è semplicissima: **pixel per FWHM = FWHM angolare / campionamento**

Sampling therefore determines how many pixels we use to describe the stellar spot produced by the sky and the optical system.

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### 8. Nyquist: why we talk about approximately 2–3 pixels per FWHM

The Nyquist criterion, in its simplified form useful for astrophotography, says that to correctly describe a structure of a given size, at least approximately two samples are needed for that dimension.

For this reason, in practice, a range of approximately 2–3 pixels per FWHM is often considered a useful reference for balanced sampling.

It is not an absolute law: the result depends on the type of object, the processing method, the signal-to-noise ratio and the actual behaviour of the system. But it is a very useful reference for understanding when we are broadly undersampled or oversampled.

### 9. Undersampling and oversampling

Undersampling means that we are describing a structure with too few pixels. If a star has an FWHM of 2.4” and the sampling is 2.4”/pixel, we have about 1 pixel per FWHM: this is too little to describe the shape of the star well.

Oversampling means the opposite: we use many more pixels than are necessary to represent the resolution actually available.

With an FWHM of 2.4” and 0.6”/pixel, we have 4 pixels per FWHM. The image is not automatically wrong, but we are recording very finely a structure whose size is largely determined by the seeing and the optical system.

Oversampling therefore does not mean a 'bad photo'. It mainly means that the pixel scale is finer than the resolution we can normally exploit.

### 10. One fundamental point: measured FWHM is not simply the seeing

This is one of the most important points in the entire discussion.

If PixInsight measures an FWHM of 2.36”, we cannot automatically conclude that the seeing was 2.36”.

The FWHM observed in the images is the combined result of several contributions:

- atmospheric seeing;
- optical quality;
- focus;
- guiding and tracking;
- possible motion blur;
- vibrations;
- collimation;
- sampling;
- possible processing effects.

It is therefore more accurate to speak of stellar FWHM or FWHM of the acquired image quality. FWHM is an excellent practical indicator of session quality, but it is not a pure, isolated measurement of seeing.

### 11. How to actually measure the FWHM at your location

There is no need to guess the seeing value; we can start by analysing real images.

With PixInsight, both SubframeSelector and WBPP can measure the FWHM and eccentricity of individual subframes.

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For a useful measurement, it is best to start with images:

- not processed;
- with unsaturated stars;
- with good focus;
- with working guiding;
- without obvious motion blur;
- preferibilmente con esposizioni abbastanza brevi.

Example with the 150/600: if SubframeSelector / WBPP return 3.12 px and the sampling is 1.59”/px:

**3.12 × 1.59 = 4.96”**

This means that the FWHM of the star measured in that subframe corresponds to about 5”. It does not automatically mean that the seeing was 5”. Focus, guiding, wind, vibrations, optics, etc. must also be considered.

It is very useful to observe FWHM + eccentricity + its trend during the night. If the FWHM goes, for example, from 4.0” to 2.4” and then returns to 3.2”, you have a concrete indication of how image quality evolved, even though you cannot attribute every variation exclusively to seeing.

### 12. The relationship between seeing, FWHM and sampling

We can now put everything into a logical chain:

**ATMOSPHERE → SEEING → resulting star size → FWHM → TELESCOPE + CAMERA → SAMPLING → number of pixels describing the star**

| Concept | What it means |
|---|---|
| Seeing | How much the atmosphere limits resolution. |
| FWHM | How wide the resulting star is in our image. |
| Sampling | How many arcseconds of sky each pixel represents. |
| Pixel/FWHM | How many pixels we use to describe the star. |

The most important sentence is: Sampling does not improve seeing. It determines how finely you record what the seeing allows you to see.

### 13. Comparing two example systems

Let us assume, purely as an example, a real FWHM of 3”.

| System | Sampling | Pixel/FWHM |
|---|---|---|
| 150/600 + ASI294 | 1.59”/px | 3 / 1,59 = 1,89 px |
| RC8 + ASI294 | 0.59”/px | 3 / 0,59 = 5,08 px |

In this example, the 150/600 uses about 2 pixels for the FWHM; the RC8 uses about 5. The RC8 has not improved the seeing: it is simply describing the same angular scale with more pixels.

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### 14. What this means in practice

With the 150/600 + ASI294 at 1.59”/px, you have a wider field, generally lower guiding-accuracy requirements, and a configuration suited to many medium-to-large subjects.

With the RC8 + ASI294 at approximately 0.59”/px, you have a much narrower field and a much finer scale. This is useful for small objects and fine detail, but makes guiding, focus, collimation and seeing issues more apparent.

The RC8 can therefore still be useful even if it is oversampled relative to average conditions: oversampling does not make the system unusable. It is simply a choice of finer scale, with corresponding costs in terms of signal per pixel, sensitivity to errors and quality requirements.

### 15. Sampling alone does not determine the field of view

Sampling and field of view are related, but they are not the same thing. Sampling tells us how much sky a single pixel sees. The total field of view depends on sampling and the number of pixels in the sensor.

The ASI294 has a resolution of 4144 × 2822 pixels. With a 600 mm telescope the field is approximately 1.84° × 1.25°. With a 1624 mm telescope it is approximately 0.73° × 0.48°.

For this reason, the 150/600 is more suited to large subjects, while the RC8 is more suited to small subjects. An intermediate focal length can occupy an intermediate position.

### 16. Focal ratio and aperture are different concepts

A common mistake is to confuse focal ratio and sampling.

The 150/600 is f/4; the RC8 at approximately 1624 mm and 203 mm aperture is about f/8. Focal ratio mainly concerns system speed and the amount of light per unit area; sampling, on the other hand, depends on focal length and pixel size.

Likewise, telescope aperture does not by itself determine sampling. Aperture affects other aspects, including light-gathering and theoretical resolving power. For sampling, focal length and pixel size are what matter.

### 17. How to use sampling to choose a focal length

Once the FWHM normally obtained at the location of interest is known, a sampling range consistent with the system can be chosen.

For example, if your sessions showed a typical FWHM of around 3”, a practical range of approximately 2–3 pixels/FWHM would correspond roughly to 1.0–1.5”/pixel.

Con la ASI294, questo porta a focali indicative:

| Desired sampling | Focal length with ASI294 |
|---|---|
| 2,0”/px | approximately 477 mm |
| 1,5”/px | approximately 636 mm |
| 1,3”/px | approximately 735 mm |
| 1,2”/px | approximately 796 mm |
| 1,0”/px | approximately 955 mm |
| 0,8”/px | approximately 1194 mm |
| 0,6”/px | approximately 1593 mm |

Therefore, a refractor of approximately 600–650 mm would bring the ASI294 to around 1.5”/px, one around 800 mm to about 1.2”/px, and one around 900–1000 mm to about 1.0”/px.

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This does not mean that one of these focal lengths is 'the best': it means that you can start from the desired angular scale and mathematically derive the focal length.

### 18. The reverse approach also works: choosing the camera

If the 150/600 is kept but, for example, 1.2”/pixel is desired, the required pixel size is approximately 3.49 µm.

Therefore, a camera with pixels around 3.5 µm would bring that telescope to that scale.

With the RC8 at approximately 1624 mm, reaching about 1.0”/pixel would instead require pixels around 7.87 µm.

This demonstrates why there is no ideal camera independently of the telescope: the same camera can behave very differently at different focal lengths.

### 19. Now BINNING enters the picture

Binning completes the picture because it changes the scale at which pixels are used, but before seeing why it is useful we need to understand what it actually does to the pixels.

Up to this point we have considered the sensor using its physical pixels one by one.

La ASI294 ha pixel fisici da 4,63 µm. I suoi campionamenti nativi sono:

| **Configuration** | **BIN 1** |
|---|---|
| 150/600 + ASI294 | 1.59”/px |
| RC8 + ASI294 | 0.59”/px |

In BIN 1, each pixel is used individually, so the sampling remains the native scale of the system.

With BIN 2 or BIN 3, several pixels are instead combined to form a single image element. The simplest case to imagine is BIN 2×2: four adjacent pixels are treated together.

Imagine a small portion of the sensor:

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | | |
| 5 | 6 | 7 | 8 | | |
| 9 | 10 | 11 | 12 | | |
| | | | | | |

In the conceptual BIN 2×2 case, four pixels 1-2-5-6 are treated as one element; 3-4-7-8 as another, and so on. In practice, instead of describing the image with each individual pixel, we describe it using groups of four pixels.

This has an immediate consequence for image scale. If we combine 2 pixels across the width and 2 along the height, the linear size of the resulting element doubles and, consequently, the angular scale doubles.

| **Configuration** | **BIN 1** | **BIN 2** |
|---|---|---|
| 150/600 + ASI294 | 1.59”/px | 3.18”/px |
| RC8 + ASI294 | 0.59”/px | 1.18”/px |

Therefore, BIN does not change the telescope's focal length and does not improve seeing. It changes how we use the pixels and, consequently, the scale at which the image is represented/acquired.

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This is where BIN connects directly to FWHM. If a star has an FWHM of 3", with the RC8 in BIN 1 at 0.59"/px, we have about 5.1 pixels per FWHM. With BIN 2 at 1.18"/px, we have about 2.5 pixels per FWHM.

BIN can therefore be seen as a tool for making sampling less fine when the native sampling is very fine. We are not creating new detail and we are not improving seeing: we are choosing to represent the image at a larger scale.

There is, however, an important qualification for the ASI294 considered here: it is a modern CMOS sensor, and we should not automatically assume that BIN 2 behaves exactly like the old hardware binning of CCDs. The actual behaviour depends on the implementation of the sensor and driver. For this reason, the old rule that "BIN 2 = four times more sensitive" should not be applied automatically.

For our sampling discussion, however, we can remember one simple thing:

**BIN 1 = scala nativa dei pixel.**
**BIN 2 = accorpamento 2×2 e, nel modello semplificato, raddoppio degli arcosecondi rappresentati da ciascun elemento risultante.**

*Now let us see why this is particularly interesting with the RC8, which goes from 0.59"/px to 1.18"/px.*

### 20. BIN 2 on the RC8: the example that makes everything clear

Let us again assume an FWHM of 3".

| RC8 + ASI294 | Sampling | Pixel/FWHM |
|---|---|---|
| BIN 1 | 0.59"/px | circa 5,1 |
| BIN 2 | 1.18"/px | circa 2,5 |

Moving from BIN 1 to BIN 2 has not improved the seeing and has not increased the telescope's resolving power. We have changed the scale at which we record/represent the image, bringing the pixels/FWHM ratio closer to the practical reference of approximately 2–3 pixels/FWHM.

This is a fundamental distinction: binning does not create detail. It can make the sampling scale more appropriate when the native scale is very fine.

### 21. BIN 2 on the 150/600: why the situation is different

On the 150/600, BIN 1 is already 1.59"/px. With an FWHM of 3" we have about 1.9 pixels/FWHM, so we are already close to the practical limit.

Con BIN 2 passeremmo a 3,18"/px e avremmo: 3 / 3,18 ≈ 0,94 pixel/FWHM. Questo è decisamente verso il sottocampionamento.

Therefore, the fact that BIN 2 is interesting on the RC8 does not mean that it should automatically be used on the 150/600 as well.

### 22. Hardware BIN and software BIN: they are not necessarily the same thing

We must distinguish binning performed during acquisition from simply resizing the image after acquisition.

If you acquire in BIN 1 at 0.59"/px and then reduce the image 2× in PixInsight, the final result may have a scale of approximately 1.18"/pixel, but the sensor originally recorded at 0.59"/px.

This is conceptually different from acquiring directly in BIN 2, even though under certain conditions the final result may look similar.

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Moreover, with modern CMOS sensors, the old CCD rules should not be applied automatically. The practical meaning of binning depends on how the sensor and driver implement the mode. BIN 2 should not simply be interpreted as: “four physical pixels magically become a single pixel that is four times more sensitive”.

### 23. BIN and FWHM: the star does not change size on the sky

Suppose that with the RC8 a star has an FWHM = 5 pixels at 0.59”/px.

La FWHM angolare è: $5 \times 0{,}59 = 2{,}95” \approx 3”$

If we represent the image at twice the scale, the same star may appear approximately 2.5 pixels wide. But the star has not physically become smaller: we are describing the same information with fewer pixels.

This helps distinguish the actual resolution of the sky from the numerical scale at which we save the image.

### 24. Binning does not make different cameras equivalent

Changing camera, changing pixel size and using BIN are three different ways of changing the scale, but they are not equivalent in every respect.

A camera with larger pixels and a camera with smaller pixels used in BIN 2 may produce similar image scales, but they may differ in sensor size, QE, read noise, dynamic range, full well, sensor technology and behaviour in different acquisition modes.

Binning is therefore an operational tool, not a substitute for proper system design.

### 25. The three sampling 'controls'

| Adjustment | Effect on sampling |
|---|---|
| Increase focal length | Sampling decreases. |
| Increase pixel size | Sampling increases. |
| Increase BIN | Sampling increases. |

This is a very useful summary: focal length, pixel size and binning allow us to change the scale, but they have different consequences for field of view, detail, signal and the overall behaviour of the system.

### 26. The practical range of your setup

| Setup | Sampling | General indication |
|---|---|---|
| ASI294 + GSO 150/600, BIN 1 | 1.59”/px | Medium-to-large nebulae and wider field |
| ASI294 + circa 700 mm | 1.36”/px | Intermediate range |
| ASI294 + circa 800 mm | 1.19”/px | Intermediate range / greater detail |
| ASI294 + RC8, BIN 1 | 0.59”/px | Very fine scale, small objects |
| ASI294 + RC8, BIN 2 | 1.18”/px | Much more balanced scale under FWHM conditions of about 2.5–3” |

These indications are not rankings of equipment. They simply describe the resulting scale and the type of use toward which each configuration tends.

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### 27. The correct procedure to use in the future

1. Start with the subject: large, medium or small?
2. Consider the actual sky at your site, not a theoretical observatory seeing value.
3. Measure FWHM on your real images with PixInsight SubframeSelector / WBPP.
4. Also check eccentricity and the FWHM trend during the night, so that every problem is not automatically attributed to seeing.
5. Use a range of about 2–3 pixels/FWHM as a practical reference, without considering it an absolute law.
6. From that sampling, derive the focal length required with the camera you own.
7. Alternatively, if you want to keep the focal length, derive the required pixel size.
8. Only then consider BIN, remembering that BIN 1 and BIN 2 may have different implications depending on the camera technology and mode.
9. Finally, also consider field of view, aperture, focal ratio, optical quality, guiding, focus, collimation, weight and mount compatibility.

### 28. The final mental map

**The entire discussion can be condensed into this sequence:**

ATMOSPHERE → SEEING → PSF/star size → FWHM → TELESCOPE + CAMERA → SAMPLING → PIXELS/FWHM → undersampling / appropriate sampling / oversampling

BIN acts on the sampling branch: by combining multiple pixels, it increases the arcseconds represented by each resulting element.

### 29. The sentence to remember

**Seeing determines how finely the sky allows you to see; FWHM measures how wide a star appears in your image; sampling determines how many arcseconds each pixel represents; the ratio between FWHM and sampling tells you how many pixels you are using to describe that star; BIN changes the scale by combining multiple pixels.**

### 30. Final application to the example equipment

With the ASI294 and the 150/600 you have about 1.59”/px. If the real FWHM of the session were 3”, you would have about 1.9 pixels/FWHM: a situation close to the practical reference.

With the same ASI294 and the RC8 at approximately 1624 mm you have about 0.59”/px. With an FWHM of 3” you would have about 5.1 pixels/FWHM: a very fine scale.

With BIN 2 on the RC8 you would have about 1.18”/px and about 2.5 pixels/FWHM: in this example, a scale much closer to the practical 2–3 pixels/FWHM range.

Therefore, the RC8 is not 'wrong' because it has 0.59”/px. It simply operates at a much finer scale. BIN 2 can be a tool for bringing that scale closer to values more appropriate to the seeing conditions and the subject.

### 31. The most useful next step

At this point, there is no longer any need to use a hypothetical seeing of 2.5” or 3” to decide on your setup.

The final verification consists of taking a real session from your ASI294, preferably with the 150/600, and reading the FWHM of the subframes in PixInsight SubframeSelector or WBPP. From those data you can derive mean and median FWHM, best and worst values, the trend during the night, eccentricity and pixels/FWHM.

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From there, the reasoning becomes completely concrete:

Real FWHM at the location → sensible sampling → BIN 1/BIN 2 comparison → focal length of a possible refractor → possible pixel size of a new camera.

In this way, sampling definitively stops being an abstract formula and becomes a practical tool for understanding and designing your setup.

### 32. Checking the calculations

In any case, calculations relating to sampling, image scale and field of view can also be checked using several tools available online. They are especially useful for quickly checking results obtained with the formulas and comparing different combinations of telescope, focal length and camera.

- **Calcolatore Campionamento Astrofotografia – Blue Journey Astro**
  https://www.bluejourneyastro.com/calcolatore-campionamento-astrofotografia/
  Un calcolatore specificamente dedicato al campionamento astrofotografico.
- **Astronomy Tools – Field of View Calculator**
  https://astronomy.tools/calculators/field_of_view/
  Permette di inserire telescopio, focale, camera, dimensione dei pixel e binning e di verificare il campo inquadrato e la combinazione complessiva del sistema.

These tools do not replace an understanding of the concepts explained in this document, but they are very useful for practical verification of calculations and for quickly experimenting with different configurations.

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